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Risk Decision Systems & Data Analytics --- Anomaly Detection, Adversarial Behavior & ML

PROFILE

Years of experience in fast-paced startup environments, designing probability-driven and ML-assisted systems for user behavior modeling, system stability, anomaly detection, and anti-abuse decision mechanisms.

Experienced in collaborating across engineering, operations, product, customer-facing and management functions in fast-changing environments.

Built ETL/ELT workflows from raw data to model-ready datasets, including data integration, cleaning, filtering, time-series granularity handling, feature preparation, and data transformation for downstream modeling.

Built highly visualized data-driven decision systems and BI dashboard-style prototypes to support rapid iteration, graph-based feature analysis, and operational decision-making.

Data Story

A user population can be viewed as an amoeba moving through a behavioral space. Its collective behavior may morph into different shapes to probe and test an online system, or shift in response to external factors that are not directly observable within the system.

I designed a detection framework capable of identifying these directional shifts at their earliest stage. In other words, when the "amoeba" begins transforming into a starfish or an elongated form, the system can promptly recognize the emerging behavioral pattern. Practical applications include time-series analysis, behavioral sequence analysis, clustering, and machine-learning-based model fitting and scoring.

CORE EXPERTISE

- Series of simultaneous Non-cooperative games for N participants (clients), finding solutions to fast-reaching converging results.
- Developed behavioural anomaly-detection pipelines using time-series analysis, the Hurst exponent, and user action sequences to identify users exhibiting anomalous behaviour.
- Feature engineering & SQL-based data architecture
- *Developed online data visualization and automated decision-making mechanisms with BI dashboard-style visual presentation, similar to Tableau and Power BI.*

TECH STACK

Programming Languages: Python, Go, SQL
Data Analysis: Time-series Analysis, k-means, Probabilistic Modeling, Graph / clustering methods
ML Engines: XGBoost, CatBoost
Tools: Gitlab, Jira, confluence, Jupyter, Spyder
Language: Chinese(native), English

EXPERIENCE

Digital Asset Market Analytics & Risk Simulation | Project Consultant

Independent Project | Sep - Oct, 2026

Developed market analytics and risk simulations across five digital assets (BTC/ ETH/ SOL/DOGE/XRP), delivering strategy backtests, interactive 2D/3D analysis tools. https://rolanthhuang.github.io/Binance_Trading_Volume_3D_Terrain_Diagram/

Material Planning & Scenario Decision-Support Simulator

Independent Project | Aug, 2026

Designed and developed a browser-based material-planning prototype using synthetic data to integrate BOMs, inventory, purchase orders, supplier ETAs, customer demand, and partial-delivery rules. The project includes static review dashboards, weekly simulation, and a rotatable 3D overview for analyzing shortages, shared-component allocation, supplier delays, short deliveries, and purchasing-expedite scenarios.
https://rolanthhuang.github.io/material-planning-scenario-simulator/outputs/material-planning-3d-order-containers.html

LDR, Online Payment & Transaction Platform — Senior Data Analyst / ML Eng (Remote)

Oct, 2025- Jan, 2026

Owned user-behavior analytics, anomaly detection, and risk decision systems for a large-scale online payment and transaction platform, spanning raw-data processing, feature engineering, model development, and real-time decision logic. The systems targeted fake traffic, bot activity, proxy or spoofed devices, and coordinated behavior while reducing manual review and false-positive interventions.

- Designed and built a two-stage anomaly-detection pipeline targeting device-based attacks, proxy/spoofed-device behavior, and abnormal user activity.
- Applied time-series features, Hurst exponent and spike-detection signals at the first-stage detection layer, achieving 80%+ recall.
- Integrated XGBoost/ CatBoost for supervised risk scoring, achieving 90%+ recall and 70%+ precision on production-labeled datasets. Applied k-means, graph-based analysis, and relationship modeling to identify coordinated and anomalous behavioral patterns across accounts and devices.
- Built multi-pipeline feature architecture to handle evolving user behavior and behavior drift.
- Integrated model outputs and detection rules into real-time decision logic, reducing manual review requirements and false-positive interventions while supporting rapid production iteration.

Anomaly detection dynamic demo: *https://rolanthhuang.github.io/hurst-spike-risk-demo/web/hurst-spike-replay.html*

https://www.dropbox.com/scl/fo/79dsnbmjs8pt1pm5hc9o9/AIxSXyfTG-m6CNhUq_2Gr50?rlkey=fe3qs7ngho3b83cgygbzayn5g&st=qtawt5tj&dl=0

JVD, Taipei (On-site) --- Senior Data Engineer

Feb, 2025 - Apr, 2025

Identified vulnerabilities of online systems. Technical documentation on product mechanisms to regulatory authorities.

Ing-fu, Taichung (On-site) --- Senior Data Engineer

Jan, 2024 - Dec, 2024

Online platform risk system algorithm from design to completion.

Online Platforms, Taipei/Taichung — Quantitative Risk & Game Systems Engineer / Mathematician (Onsite/Hybrid/Remote)

Jan, 2018 - May, 2022 / Jun, 2023 - Dec, 2023

Worked across multiple online gaming platforms, studios, and project-based teams over ~4.5 years, focusing on probability modeling, risk control, and automated decision systems in highly adversarial environments. Despite the fragmented organizational settings, the underlying role and problem domain remained consistent: designing systems that maintain long-term profitability and stability under strategic player behavior.

- Designed automated platform control systems for adversarial user environments
- Implemented algorithm-driven bot strategies under probabilistic constraints
- Led and collaborated with risk/data teams on technical direction across multiple studios.
- Built highly visual decision-support systems using graph-based features.
- Led cross-functional initiatives across mathematics and risk control, defining system design direction and establishing internal design and validation workflows used by dozens of game platforms and merchants. Operated effectively in environments with incomplete data, noisy and shifting user behavior, and adversarial system dynamics. The core design framework developed during this period became a foundation for later roles.
- Delivered actionable real-time insights to senior leadership, supporting faster and more informed business decisions.

Online Risk control system for multi-platforms/ balancing system for multi-agents:
https://rolanthhuang.github.io/realtime-multi-customer-balancing/outputs/profit_control_realtime.html
https://www.dropbox.com/scl/fo/bbqcu8bydudmingg61rkw/AIO8SNby0HozL2r--bJvTzg?rlkey=b3o601ut4iuaxniwyga8lvmdj&st=lzz86dq4&dl=0

Quanta Computers, Shanghai & California — Test Engineer

Feb, 2016 - Apr, 2017

In server assembly and shipping sites in both Shanghai and California factories. This experience involved extensive cross-functional collaboration across R&D, PCB/layout-related teams, assembly and testing teams, and shipment teams. I also worked with engineers from different countries to troubleshoot and resolve issues.

EDUCATION

Undergraduate studies in Chemistry, NCKU — transferred to NTU

2000 - 2001

B.S. Mathematics, National Taiwan University

2003 - 2008

Institute of Physics, National Taiwan University — Master (degree not completed)

2008 - 2010